

Data Modelling Architectures

Data modelling architectures provide the **framework for storing, processing, and analysing** large volumes of data in a structured and efficient manner. With the increasing complexity and scale of modern data environments, choosing the right architecture is essential for optimising data quality, performance, and accessibility. These architectures determine how data is organised, how it flows through systems, and how it can be utilised for analytics and decision-making.

Learning Objectives

1. Explore the concept of a lakehouse architecture, blending data lake flexibility with data warehouse management features.
2. Learn about the medallion architecture for processing and organising data.

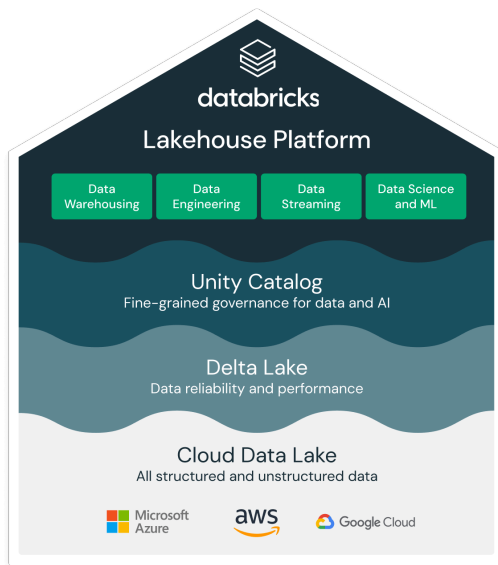
Exploring Data Modelling Architectures

Let's delve into three different data modelling architectures, understanding their principles, strengths, and applications.

Lakehouse Architecture

Lakehouse architecture is a modern data platform that combines the best features of two traditional systems: data lakes and data warehouses. A data lake is a system used to store large amounts of raw data, which can come in many different forms, such as images, videos, or logs. In contrast, a data warehouse is used to store structured and organised data, like spreadsheets or tables, which are ready for analysis and reporting. The challenge with these two systems is that they are usually separate, meaning data needs to be moved and processed between them, which can be inefficient, costly, and slow down the time it takes to get useful insights.

Lakehouse architecture overcomes this challenge by allowing both structured and unstructured data to coexist in the same environment. It combines the scalability and flexibility of data lakes, which can handle massive volumes of raw data, with the transactional capabilities and query performance of data warehouses. This architecture enables real-time analytics, advanced data processing, and machine learning directly on the raw data without the need for complex data transfers between systems.



- **Core Functionalities:** Supports data warehousing for analytics, data engineering for pipeline management, real-time data streaming, and data science/ML for advanced insights, all within the same platform.
- **Unity Catalog:** Provides fine-grained governance for secure data and AI management.
- **Delta Lake:** Ensures data reliability and performance with features like versioning and ACID transactions.
- **Cloud Data Lake:** Stores structured and unstructured data across cloud providers (Azure, AWS, Google Cloud).

Benefits of lakehouse architecture

- **Unified data platform:** Lakehouse architecture provides a single platform that stores both structured and unstructured data together. This allows data from multiple sources to be accessed, managed, and queried from one system, streamlining the process of working with diverse datasets.
- **ACID transactions:** Unlike traditional data lakes that lack robust data governance and integrity, lakehouse systems support ACID (Atomicity, Consistency, Isolation, Durability) transactions. This ensures that all data operations are reliable, and the data remains consistent even as it is processed, queried, or updated by multiple users or systems simultaneously.
- **Advanced analytics:** Lakehouse platforms enable organisations to perform advanced analytics, such as machine learning and real-time data processing, directly on the raw or semi-structured data without having to move it to a separate analytics platform. This is particularly beneficial for businesses that need to derive insights quickly from large and complex datasets.

Example

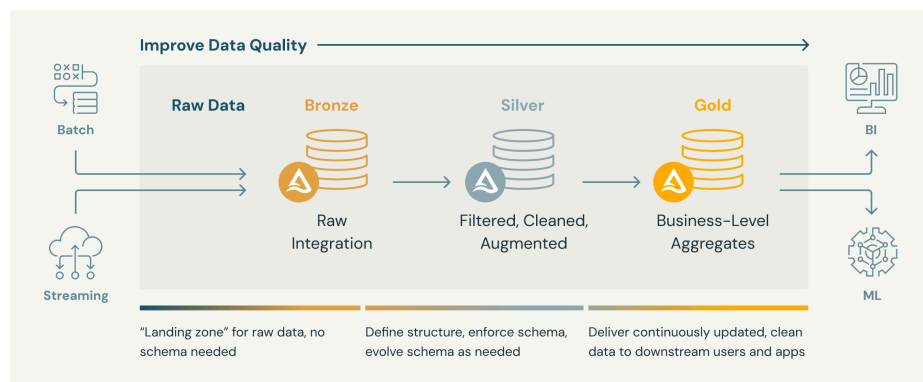
Research institutions may use lakehouse architecture to store and analyse genomic data alongside patient records, developing personalised treatment plans.

Medallion Architecture

Medallion architecture is a layered approach to organising data. It breaks down data processing into distinct stages, or layers, to gradually improve data quality and streamline processing efficiency. This structure is especially useful in environments where data is ingested continuously, such as streaming data from IoT devices, financial transactions, or real-time analytics.

Medallion architecture layers

- **Bronze layer:** The first stage where raw, ingested data is stored in its original form. This layer holds unprocessed data, which may contain duplicates, errors, or incomplete records, but is crucial for keeping the full history of the incoming data.
- **Silver layer:** In this layer, the raw data is cleaned, enriched, and transformed into a more structured form. The goal is to remove errors, apply necessary corrections, and enhance the data for easier analysis, making it more usable for broader reporting purposes.
- **Gold layer:** The final layer where data is aggregated and refined into business-level insights. Data in the gold layer is ready for high-level analysis, reporting, and decision-making, providing a reliable foundation for dashboards and machine learning models.



Benefits of medallion architecture

- **Data quality:** Incrementally improves data through each layer.
- **Efficient processing:** Streamlines data transformation process, breaking it into manageable steps.
- **Enhanced governance:** The layered structure provides clear data lineage, which means it's easy to track how data has been processed and transformed. This transparency improves governance and control.

Example

Companies may use medallion architecture to process streaming data from IoT devices, ensuring timely and accurate analysis.

When to Select which Architecture?

Selecting the right data modelling architecture—whether Lakehouse, Medallion, or Data Vault—depends on the specific needs of our project, such as data types, scale, and processing requirements. By aligning the architecture with our organisation's goals, we can optimise data quality, performance, and scalability, ensuring effective data management and analysis.

Before flipping the cards, take a moment to consider which architecture you would choose based on the following requirements:

Flexibility and advanced analytics

Opt for lakehouse architecture when dealing with varied data types

Streamlined data processing

Apply medallion architecture for efficient handling of streaming data.

Reflect:

An organisation wants to predict and prevent the spread of infectious diseases using diverse data sources. Which data architecture would you recommend and why?

Conclusion

Choosing the right data modelling architecture is critical for managing and leveraging large, complex datasets effectively. Lakehouse architecture offers flexibility by combining structured and unstructured data in one platform, supporting advanced analytics and real-time insights. Medallion architecture provides a clear, layered process to improve data quality and streamline workflows, especially in streaming environments. The best choice depends on the organisation's specific data types, scale, and analytical needs.

Key Takeaways

- Takeaway 1: Lakehouse architecture unifies structured and unstructured data, enabling advanced analytics without complex data transfers.
- Takeaway 2: Medallion architecture improves data quality through bronze, silver, and gold layers, supporting efficient processing and analysis.
- Takeaway 3: ACID transactions in lakehouse systems ensure data integrity and reliability during multi-user operations.
- Takeaway 4: Architecture selection should align with project goals, data types, and performance requirements for optimal results.